

# Hershdeep Singh Chopra (Hersh)

Hershdeepchopra2026@u.northwestern.edu | (424) 832-0952  
<https://hersh42.github.io/>

## Education

---

**Northwestern University**  
Ph.D. in Economics

Expected June 2026

**University of California, Los Angeles (UCLA)**  
Bachelor of Science in Mathematics and Economics (with honors)  
Cumulative GPA: **3.8/4.0**

2016 – 2020

## References

---

Professor Jeffrey Ely  
Northwestern University  
[jeffely@northwestern.edu](mailto:jeffely@northwestern.edu)

Professor Bruno Strulovici  
Northwestern University  
[b-strulovici@northwestern.edu](mailto:b-strulovici@northwestern.edu)

Professor Alessandro Pavan  
Northwestern University  
[alepavan@northwestern.edu](mailto:alepavan@northwestern.edu)

## Research Interests

---

Game Theory, Information Design, Mechanism Design, Statistical Decision Theory

## Job Market Paper

---

### **Employer Competition and Certification**

**Abstract:** This paper develops a theory of employer competition over hiring standards in labor markets where employers rely on third-party certification to screen applicants. The certifier sells tests to an applicant, who possesses noisy private information about his ability and seeks to persuade employers to offer him employment. An employer hires if, conditional on the test outcome, her posterior belief about the applicant's ability meets her standard. Employers compete in their selectivity or hiring standards to attract the applicant. The certifier faces both a screening problem, due to applicants' private information, and an information design problem, as tests must be informative enough to persuade employers. The screening frictions faced by the certifier affect the equilibrium determination of hiring standards and the certification mechanism. I show that this inefficiency in test allocation leads to two distortions relative to a benchmark where the certifier only faces an information design problem. First, there is an increase in the probability of unemployment. The certifier's optimal selling mechanism leads to the exclusion of some applicant types. Second, competition among employers intensifies. The inefficient test allocation steers the applicant supply towards less selective employers, resulting in equilibrium constriction of standards.

## Working Papers

---

### **Incentive-Compatible Information Design** (joint with Jeffrey Ely)

**Abstract:** We study the design of mechanisms by an intermediary that generates information for a sender to persuade a receiver about an unknown attribute of the sender. The sender is initially privately, but imperfectly, informed about her attribute, and the receiver takes an action based on posterior beliefs about the sender's attribute and the sender's belief about the attribute. The design of the mechanism, therefore, confronts both incentive-compatibility constraints (for the sender) and obedience constraints (for the receiver). We characterize profit-maximizing mechanisms when the intermediary contracts with the sender, and we specialize to two applications: the design of college-admissions tests and the optimal use of consumer data on a digital market platform. We also show how our analysis extends beyond a profit-maximizing intermediary by analyzing the design of budget-balanced selling mechanisms in a lemons market.

**Interim Information Design** (joint with Jeffrey Ely)

**Abstract:** We study efficient and consumer-surplus maximizing information policies in a bilateral trade setting where the buyer is initially privately imperfectly informed about his willingness to pay. We show that providing more information to the buyer can lead to higher market prices without affecting the consumer or producer surplus.

**TEACHING EXPERIENCE**

---

**Teaching Assistant, Northwestern University**

|                                                                                    |                                     |
|------------------------------------------------------------------------------------|-------------------------------------|
| Econ 331: Economics of Risk and Uncertainty (under Professor Marciano Siniscalchi) | Fall 2024                           |
| Econ 380: Game Theory (under Professor Asher Wolinsky)                             | Fall 2023                           |
| Econ 313: Economics of Data (under Professor Annie Liang)                          | Spring 2023                         |
| Econ 310-2: Intermediate Microeconomics (various faculty members)                  | Winter 2024, Spring 2022, Fall 2021 |

**RESEARCH EXPERIENCE**

---

|                                                                                     |             |
|-------------------------------------------------------------------------------------|-------------|
| Research Assistant, Professor Jeffrey Ely, Northwestern University                  | 2024 – 2025 |
| Research Assistant, Professor Annie Liang, Northwestern University                  | 2022        |
| Research Assistant, Professor Simon Board and Professor Moritz Meyer-ter-Vehn, UCLA | 2018 – 2019 |